09 ERGO-CONTOUR

Palm swell, four finger flutes and a thumb rest for hand conformance.

64.1 g

PLA · 100 % infill · sealed core

✓ LIGHTENED CORE · SLOT + HOLE



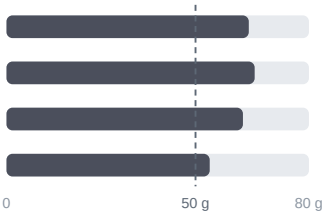
3/4 view · rendered from the STL, colours are preview only

AT A GLANCE

Overall height	**135.00 mm**
Max footprint (foot)	**42.63 × 36.61 mm**
Paddle slot	**30.2 × 12.2 mm, 130.0 deep**
Textured band	**z 21.5 – 116.0 mm (94.5 tall)**
Outer size in band (max)	**38.0 × 30.9 mm**
Peak texture height	**2.98 mm above base**
Screw hole (butt)	**Ø 3.5 mm + Ø 6.2 × 2.2 counterbore**
Material volume / surface	**51.69 cm³ / 433 cm²**

WEIGHT · 100 % INFILL

PLA	1.24 g/cm³	**64.1 g**
PETG	1.27 g/cm³	**65.6 g**
TPU	1.21 g/cm³	**62.5 g**
ABS	1.04 g/cm³	**53.8 g**



COLOURWAY

#424248

3MF print palette

PRINT PROFILE

Infill / layer height	**100 % / 0.20 mm**
Layers	**675**
Orientation	**upright, butt end down**
Supports	**none required**
Sealed hollow core	**35.5 cm³, walls ≈ 1.2 mm**
Filament, 1.75 mm	**≈ 21.5 m (100 % flow)**

DESIGN INTENT

A smooth grip shaped for the hand rather than textured: an hourglass swell, a raised palm pad on the back, four finger flutes and a thumb dish on the front. There are no sharp edges anywhere on the surface.

The smoothest surface in the set; it prints with the least visible layer texture.

KEY FEATURES

- Four finger flutes at z = 44, 58, 72, 86 mm
- Thumb dish at z about 97 mm, palm pad at z about 62 mm
- Hourglass swell 1.73 – 2.33 mm, peaking at about 2.8 mm on the palm pad

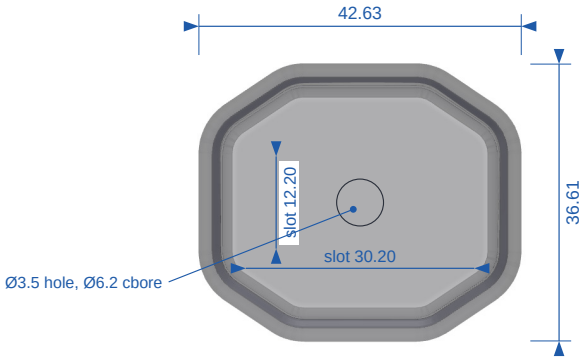
VERIFIED ON THE STL

Paddle slot	**30.2 × 12.2 mm (30 × 12 + 0.2)**
Screw hole + counterbore	**Ø 3.5 mm + Ø 6.2 × 2.2**
Outer envelope vs reference	**+0.000 mm at most**
Outer body / sealed cavity	**1 / 1**
Wall around the cavity (min)	**1.11 mm**
Material under 0.4 mm wide	**0.014 % of volume**
Widest flat bridge	**3.6 mm**

FILES

stl/09_ergo_contour.stl	5.0 MB · 100,336 triangles
3mf/09_ergo_contour.3mf	1.1 MB · colour

✓ SLOT & HOLE VERIFIED ✓ ONE BODY · SEALED CORE ✓ PRINT-READY · 0.20 MM LAYERS ✓ 3MF VALIDATED

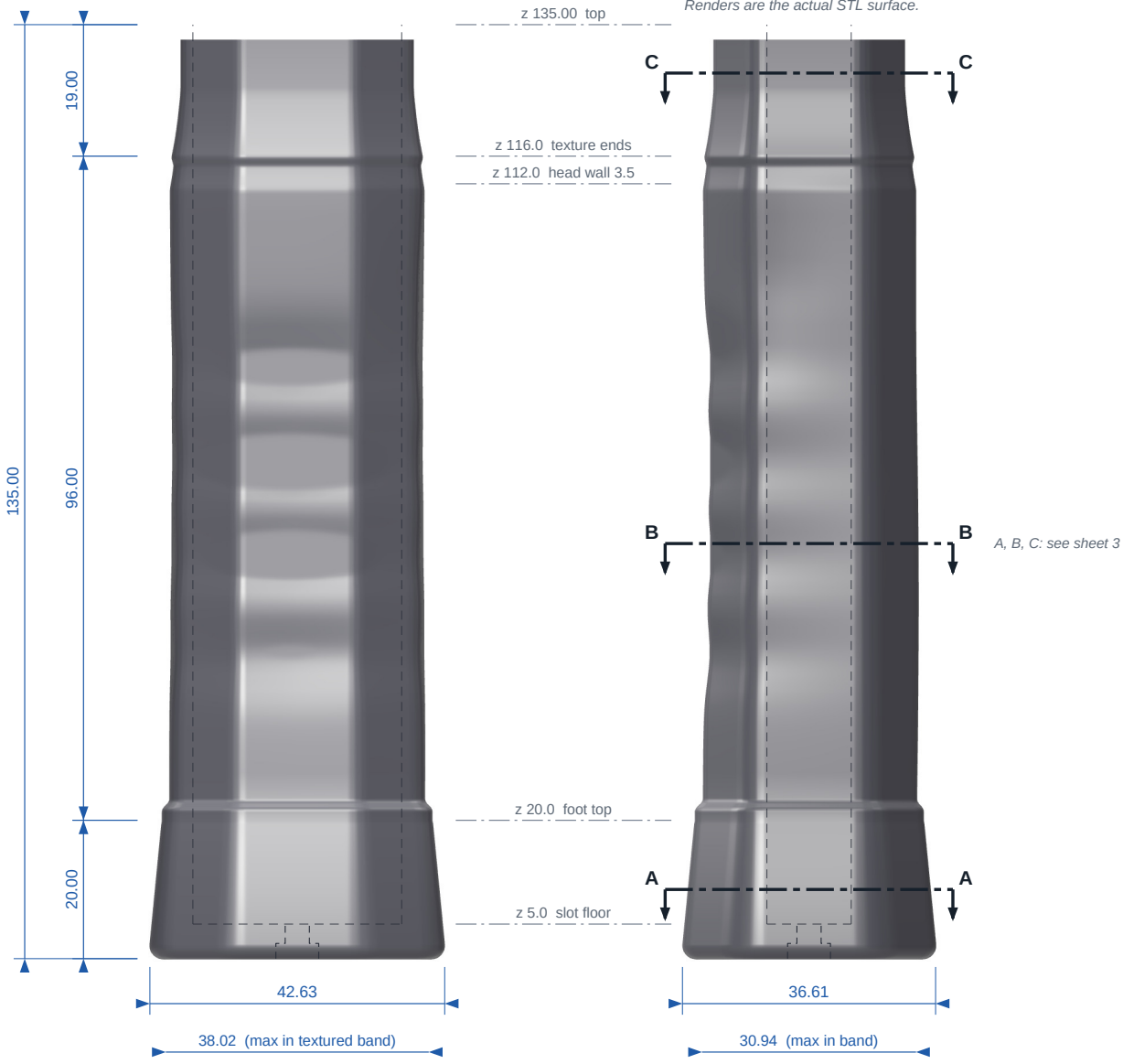


TOP VIEW (from +Z)

KEY DIMENSIONS

Overall height	**135.00**
Overall width × depth (foot)	**42.63 × 36.61**
Paddle slot width × depth	**30.20 × 12.20**
Slot depth (floor at z 5.0)	**130.00**
Outer W × D, textured band (max)	**38.02 × 30.94**
Foot top (z 20)	**39.15 × 33.13**
Butt face (z 0)	**40.56 × 34.54**
Peak texture height above base	**2.98**
Screw hole / counterbore	**Ø 3.50 / Ø 6.20 × 2.2**

All values in mm. Dashed lines: paddle slot and screw hole (hidden edges). Renders are the actual STL surface.



FRONT VIEW (from −Y)

RIGHT VIEW (from +X)

SCALE 1 : 1



PART

ERGO-CONTOUR

Pickleball paddle grip sleeve, closed butt, open throat

GENERAL NOTES

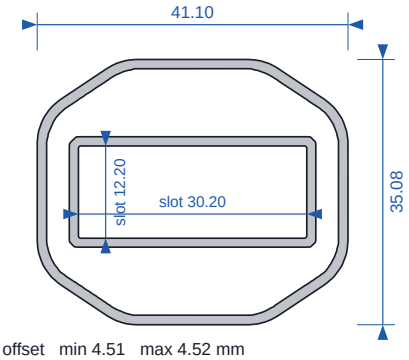
1. Views are true size: print this sheet at 100 %, do not fit to page.
2. Dimensions are measured from the STL, not nominal.
3. Sealed hollow core, walls ≈ 1.2 mm; head above z 100 solid, end walls slit 20 mm.
4. Outer envelope matches PickleballGrip_1.3mf below z 90; head thickened.

SCALE **1 : 1**	UNITS **millimetres**	PROJECTION
MATERIAL **PLA / PETG / TPU**	INFILL **100 %**	LAYER HEIGHT **0.20 mm**
MASS (PLA) **64.1 g**	VOLUME **51.69 cm³**	SURFACE AREA **433 cm²**
REVISION **7**	DATE **2026-10-22**	SHEET **2 of 4**

Sections & profiles

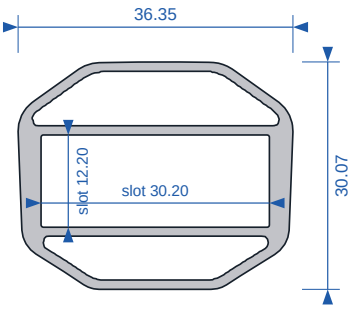
True-size cross-sections, how the outer surface and material change with height, and the dimension tables.

SECTION A–A z = 10 mm · foot



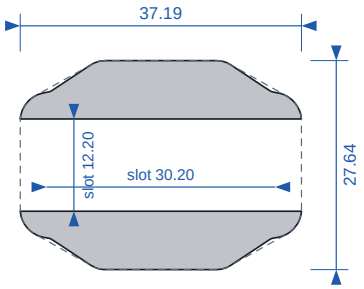
Skin offset min 4.51 max 4.52 mm
Area 268 mm² · outer perimeter 128.9 mm

SECTION B–B z = 60 mm · textured band



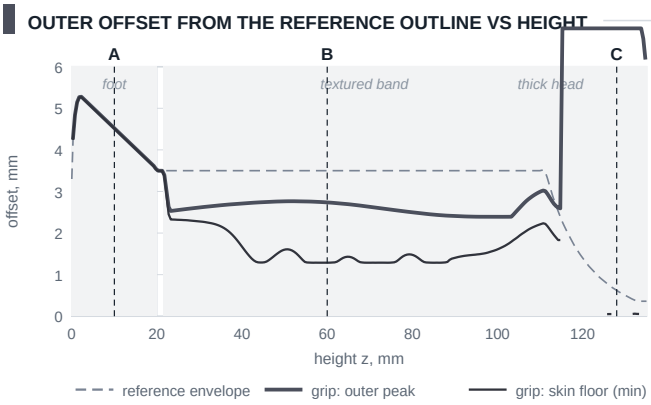
Skin offset min 1.29 max 2.74 mm
Area 260 mm² · outer perimeter 112.6 mm

SECTION C–C z = 128 mm · rim

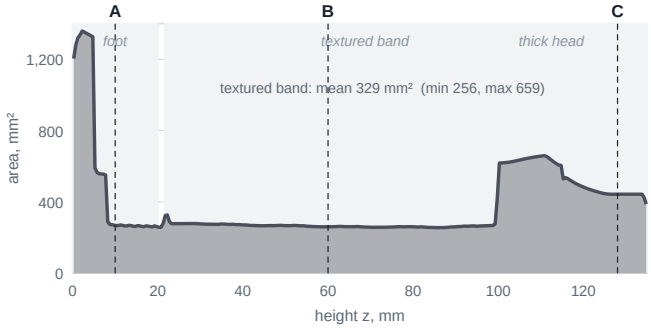


Skin offset max 6.92 mm
Area 443 mm² · outer perimeter 110.7 mm

OUTER OFFSET FROM THE REFERENCE OUTLINE VS HEIGHT



MATERIAL CROSS-SECTION AREA VS HEIGHT



OVERALL & SLOT

Overall length (Z)	**135.00 mm**
Overall width (X)	**42.63 mm**
Overall depth (Y)	**36.61 mm**
Slot width × depth	**30.20 × 12.20 mm**
Slot area / perimeter	**368.4 mm² / 84.3 mm**
Slot depth (floor at z 5.0)	**130.00 mm**
Screw hole / counterbore (butt)	**Ø 3.50 / Ø 6.20 × 2.2**
Sealed cavity / min wall	**35.5 cm³ / 1.11 mm**
Outer W × D in band (max)	**38.02 × 30.94 mm**
Material volume / surface	**51.69 cm³ / 432.7 cm²**

ZONES BY HEIGHT

ZONE	Z FROM	Z TO	HEIGHT
Solid floor (butt)	0.0	5.0	**5.0**
Flared foot	0.0	20.0	**20.0**
Smooth collar	20.0	21.5	**1.5**
Textured band	21.5	116.0	**94.5**
Smooth neck	116.0	112.0	**-4.0**
Head (solid, thick wall, slot opening)	112.0	135.00	**23.0**

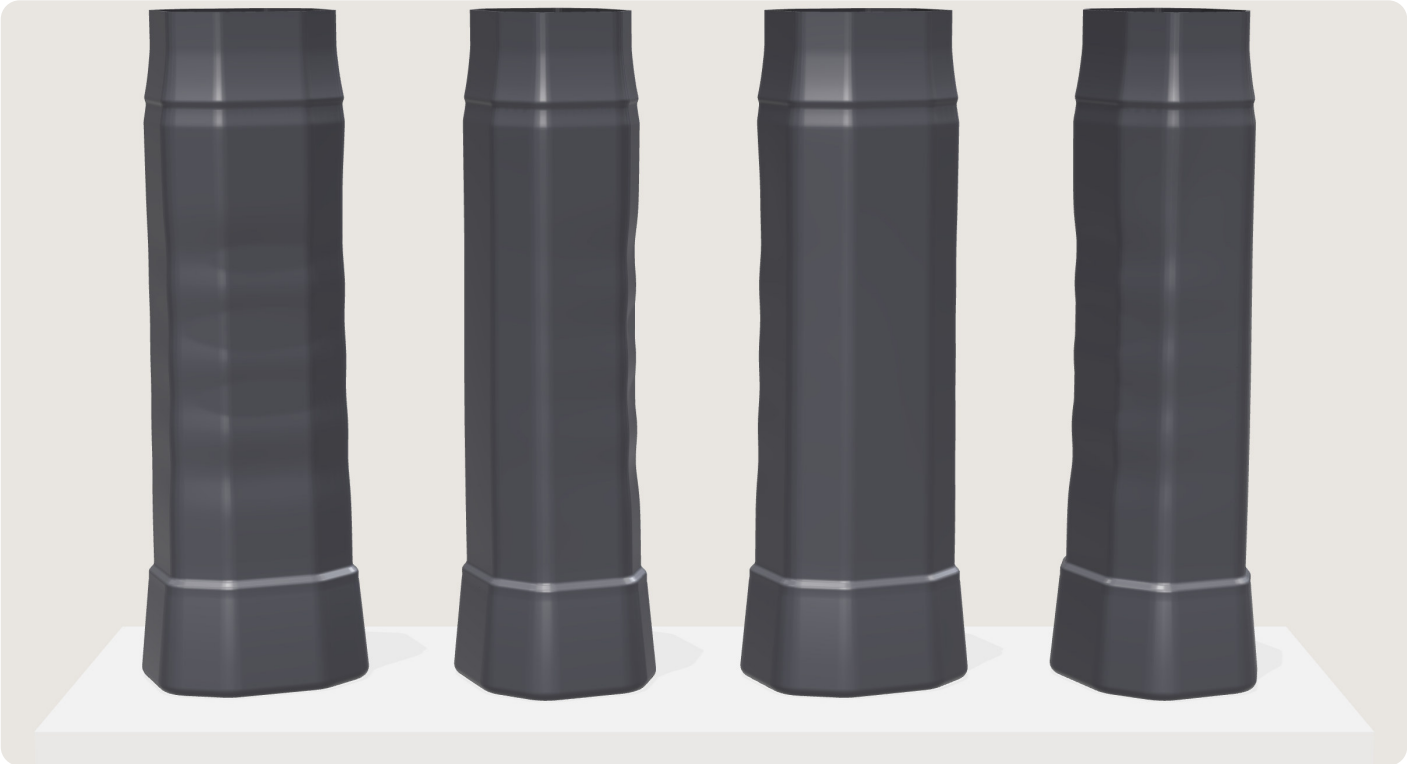
Relief floor above reference line (design / min)	**1.20 / 1.29 mm**
Slot wall over the head (design / min)	**3.50 / 3.55 mm**
Peak texture height above base	**2.98 mm**

OUTER DIMENSIONS BY HEIGHT

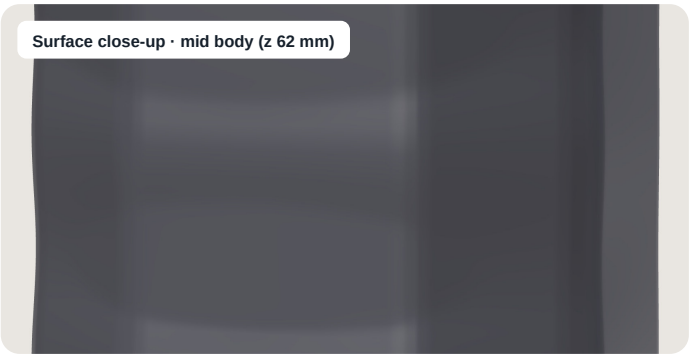
Z (MM)	OUTER W	OUTER D	PERIMETER	AREA MM²	SKIN MIN	PEAK OFFSET
5.0	42.08	36.06	132.0	959	5.00	5.01
10.0	41.10	35.08	128.9	269	4.51	4.52
20.0	39.11	33.09	122.6	260	3.52	3.52
30.0	36.93	30.94	115.6	277	2.28	2.62
45.0	36.60	30.16	113.3	266	1.30	2.74
60.0	36.35	30.07	112.6	261	1.29	2.74
75.0	36.18	29.90	112.2	259	1.29	2.56
90.0	36.27	30.01	112.5	261	1.40	2.41
105.0	37.21	30.69	114.9	635	1.92	2.59
116.0	37.20	30.19	114.2	534	open	6.92
125.0	37.19	27.75	110.8	447	open	6.92
132.0	37.19	27.64	110.7	443	0.06	6.92

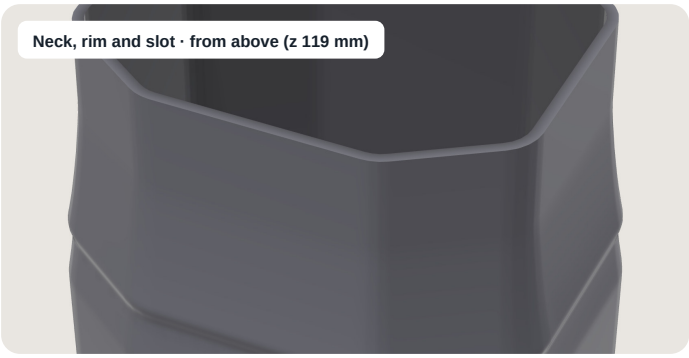
Views & surface

Four-way turntable, close-ups of the surface, and the unrolled height map with the design parameters.

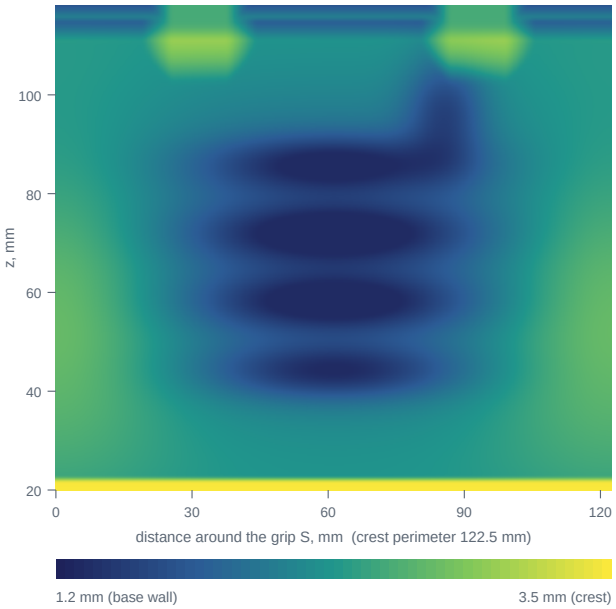


Four views, 90° apart · rendered from the STL, colours are preview only





UNROLLED SURFACE · HEIGHT ABOVE BASE



DESIGN PARAMETERS

Finger flutes	**4, at z 44 / 58 / 72 / 86**
Flute width / depth	**about 9 mm / 0.8 mm**
Thumb dish	**z 97 mm, depth 0.6 mm**
Palm pad	**back face, z 62 mm, +1.0 mm**
Hourglass swell	**1.73 – 2.33 mm above base**
Peak height	**about 2.8 mm above base**
Surface	**smooth, no sharp edges**

PRINT & VERIFICATION

Result: slot & hole / print-ready	**PASS / PASS**
Outer body · genus · sealed cavity	**1 · 1 · 1**
Wall around the cavity (min)	**1.11 mm**
Paddle slot	**30.2 × 12.2 mm**
Envelope over reference	**+0.000 mm**
Material under 0.4 mm wide	**0.014 %**
Widest flat bridge	**3.6 mm**